

Technical Datasheet Input / Output Modules with MQTT Protocol with Ethernet Interface

The **IO Module** communicates via an **Ethernet Interface** and is designed to interface seamlessly with other **MQTT-compatible devices**. The module acts as an **input/output device** that bridges the **MQTT network** and the connected gateway, enabling real-time data exchange over the MQTT protocol. This **Digital I/O module** is robust, energy-efficient, and user-friendly, making it ideal for a wide range of IoT applications. It supports both **digital inputs** and **digital outputs** and allows users to easily publish and subscribe to topics on an MQTT broker, facilitating communication with other devices, cloud platforms, and remote monitoring systems.

4 Port AI Module with MQTT-Ethernet: -



The IO modules are mounted on DIN rail mountable casing and with exposed connectors and LED indicators.

The design of the modules incorporates '**resettable Fuses**' to safeguard against sudden High Voltage spikes. The board also has '**reverse polarity**' protection both for input **Power** connector.

The communication port also has signal isolation and additional Jumper for 'end of line' 120 ohms termination resistance which may or may not be used.

Specifications

General:-

I/O Connectors	2 Pin 5.08 mm pitch pluggable screw terminals.
Dimensions	70 mm L x 110 mm B x 50 mm H
Power	Input Power – 12-24 VDC Typical – 12V DC @ 1A
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



AI Inputs:-

Channels	4
Input Signal	4 – 20 mA / 0 – 10VDC
Accuracy	± 2% Full scale
Input Resolution	10 bits / 12 bits (Optional)
Isolation	Optically Isolated
External Loop Voltage	+ 12 VDC min
AI input impedance	120Ω

Technical Specifications:-

- **MQTT Protocol Support:** Fully compliant with the MQTT protocol, supporting both MQTT 3.1.1 and MQTT 5.0.
- **Ethernet Connectivity:** Standard Ethernet port (10/100 Mbps) for stable and high-speed data transmission.
- **Low Power Consumption:** Optimized for low power usage, ideal for battery-operated or energy-efficient applications.
- **Easy Integration:** Supports standard libraries and tools for easy integration with existing systems.
- **Message Retention & Quality of Service (Q o S):** Supports various Q o S levels (0, 1, 2) to guarantee message delivery.

- **Secure Communication:** Support for TLS/SSL encryption for secure communication with MQTT brokers.
- **Real-time Data Handling:** Efficient, real-time messaging with minimal delay.
- **Compact Form Factor:** Small and lightweight, designed for embedded systems.

To set up MQTT over Ethernet and find a device's IP address using its MAC address, you'll follow these general steps. This involves setting up an MQTT broker (e.g., Mosquitto, MQTTX, or another broker), connecting devices to the network, and using the MAC address to find the device's IP. Below is a detailed procedure for setting up MQTT with Ethernet and using a MAC address to find the IP address of a device.

- **How To Find the IP Address using MAC Address.**

1. **Open an IP Scanner Tool**

Open an IP scanner tool like **Advanced IP Scanner** on your device. You can also use any other IP scanner software if you prefer.

2. **Scan Your Network**

Click the "Scan" button to scan the devices connected to your local network. The scanner will list all devices, showing their IP addresses and MAC addresses.

3. **Locate the Device by MAC Address**

Look for the device using the **MAC Address** that was provided with the device. The MAC address will be unique to that device, so it will help you identify it.

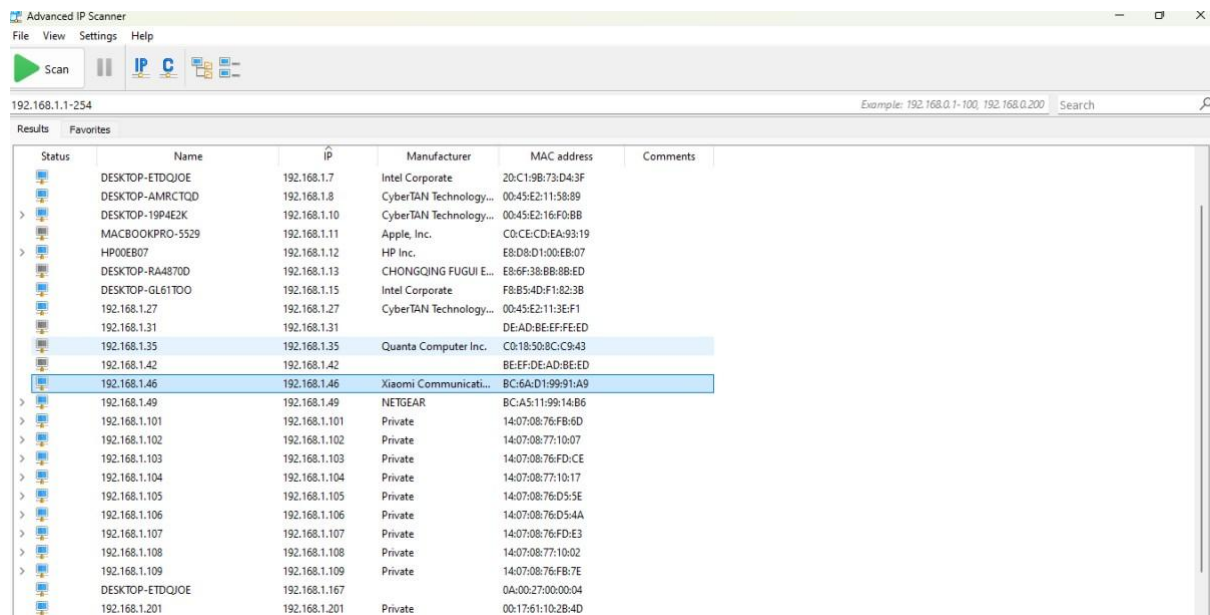
4. **Find the IP Address**

Once you locate the device by its MAC address, you will see the **IP Address** next to it. This is the IP address assigned to the device on your network.

5. **Use the IP Address**

Enter this IP address in your any Internet Browser or network configuration to connect to the device.

Note: You can refer to the **image** provided, where the **MAC address** and its respective **IP address** are clearly listed.



Status	Name	IP	Manufacturer	MAC address	Comments
	DESKTOP-ETDQJOE	192.168.1.7	Intel Corporate	20:C1:9B:73:D4:3F	
	DESKTOP-AMRCTQD	192.168.1.8	CyberTAN Technology...	00:45:E2:11:58:89	
	DESKTOP-19P4EZK	192.168.1.10	CyberTAN Technology...	00:45:E2:16:F0:BB	
	MACBOOKPRO-5529	192.168.1.11	Apple, Inc.	C0:CE:CD:EA:93:19	
	HP00EB07	192.168.1.12	HP Inc.	E8:D8:D1:00:E8:07	
	DESKTOP-RA4870D	192.168.1.13	CHONGQING FUGUI E...	E8:6F:38:BB:8B:E0	
	DESKTOP-GL61TDO	192.168.1.15	Intel Corporate	F8:B5:4D:F1:82:3B	
	192.168.1.27	192.168.1.27	CyberTAN Technology...	00:45:E2:11:3E:F1	
	192.168.1.31	192.168.1.31		DE:AD:BE:EF:FE:ED	
	192.168.1.35	192.168.1.35	Quanta Computer Inc.	C0:18:50:8C:C9:43	
	192.168.1.42	192.168.1.42		BE:EF:DE:AD:BE:ED	
	192.168.1.46	192.168.1.46	Xiaomi Communicati...	BC:6A:D1:99:91:A9	
	192.168.1.49	192.168.1.49	NETGEAR	BC:A5:11:99:14:B6	
	192.168.1.101	192.168.1.101	Private	14:07:08:76:FB:6D	
	192.168.1.102	192.168.1.102	Private	14:07:08:77:10:07	
	192.168.1.103	192.168.1.103	Private	14:07:08:76:FD:CE	
	192.168.1.104	192.168.1.104	Private	14:07:08:77:10:17	
	192.168.1.105	192.168.1.105	Private	14:07:08:76:D5:5E	
	192.168.1.106	192.168.1.106	Private	14:07:08:76:D5:4A	
	192.168.1.107	192.168.1.107	Private	14:07:08:76:FD:E3	
	192.168.1.108	192.168.1.108	Private	14:07:08:77:10:02	
	192.168.1.109	192.168.1.109	Private	14:07:08:76:FB:7E	
	DESKTOP-ETDQJOE	192.168.1.167		0A:00:27:00:00:04	
	192.168.1.201	192.168.1.201	Private	00:17:61:10:28:4D	

- After Entering the ip You will see interface as shown in the image.

Note: - (For more help you can refer below image1.1 for login)

Configuration Center

Network Settings

IP Address:

Subnet Mask:

Gateway:

DNS Server:

Save Network

MQTT Configuration

Host IP or Domain:

Topic:

Subscriber:

Username:

Password:

Port:

Data Rate/Sec:

Save MQTT

Image (1.1)

Configuration Centre

Network Settings:-

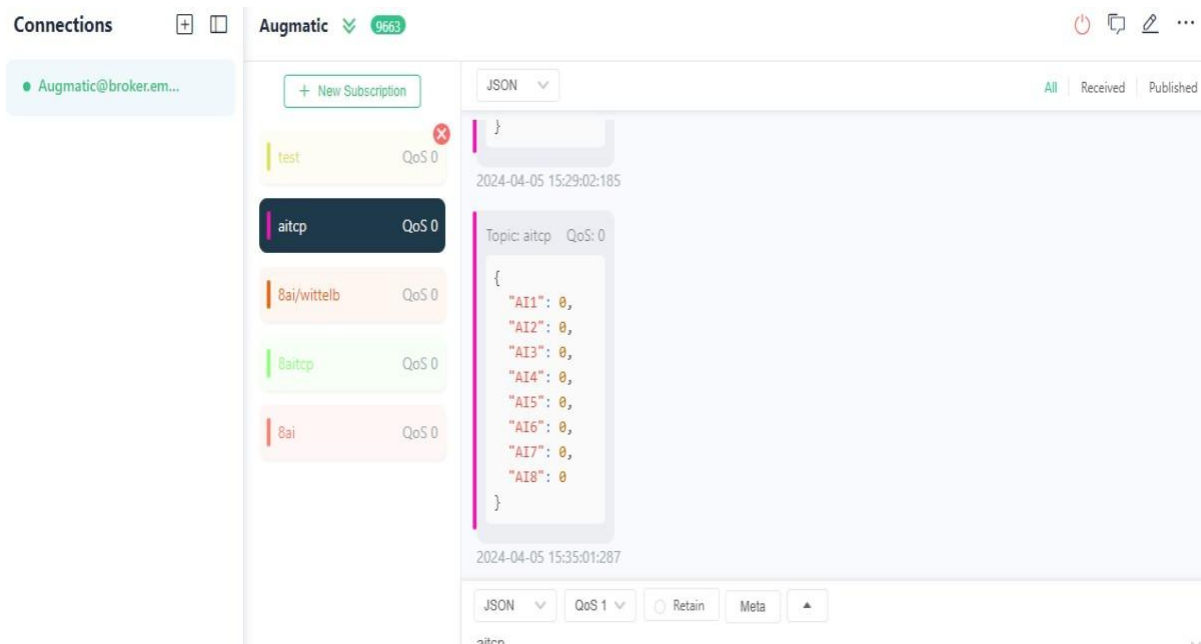
- **IP Address :** (Example:192.168.1.254)
Assign a valid IP Address in your LAN range.
- **Subnet Mask:** (Example:255.255.255.0)
Usually 255.255.255.0 for most offices and home networks.
- **Gateway:** (Example: 192.168.1.1)
Enter your routers IP Address.
- **DNS Server:** (Example: 8.8.8.8)
Google DNS or your local DNS server.
- Click **Save Network** to Apply Settings. The device will restart with new network Configuration.

MQTT Configuration:-

- **Login Prompt**
When you open the device's details page, you will be prompted to enter a **username** and **password**.
- **Default Username**
The default **username** is usually admin.
- **Default Password**
The default **password** is typically admin or password.
- **MQTT Port**
The default MQTT port is **1883** for unencrypted connections.
For secure MQTT over TLS, use **port 8883**.
- **Topic Name**
Ensure you configure the correct **topic name** where digital i/o data will be published.
For example, if you set the topic to "aitcp", you will receive analog and digital data on that topic.
- **Subscriber:Topic where device will listen for commands to operate relays.**
- **Data Rate:**
The **data rate** refers to how often the device publishes messages. You can set it to a specific interval, such as every second or every few seconds, depending on how often you want the data to update.

Note: (Username Password is optional).

- Click **Save MQTT** to store MQTT Settings. The device will attempt to connect to the broker with provided details
- **Factory Reset** : If you forgot the device IP or Enter the incorrect Network /Mqtt Settings you can restore the device to factory defaults using Reset Jumper.
- **Steps to reset:**
 1. Power ON the device
 2. Place Reset Jumper in Reset Position
 3. Press RESET button
 4. Remove jumper
 5. Power On Device
- Settings will be restored



- Here you can see the data of **analog input** in **JSON format**, which displays the real-time analog values from all input channels.

Analog Input Data via MQTT:-

To monitor analog input values through an MQTT cloud system, the device publishes real-time analog data to the configured MQTT topic. The data is sent in **JSON format**, allowing easy integration with dashboards or cloud applications.

Operation Process:

1.Publishing to Topic:

The device publishes analog input data to an MQTT topic (e.g., *wittelb_pub*) at the configured data rate.

2.Response Format:

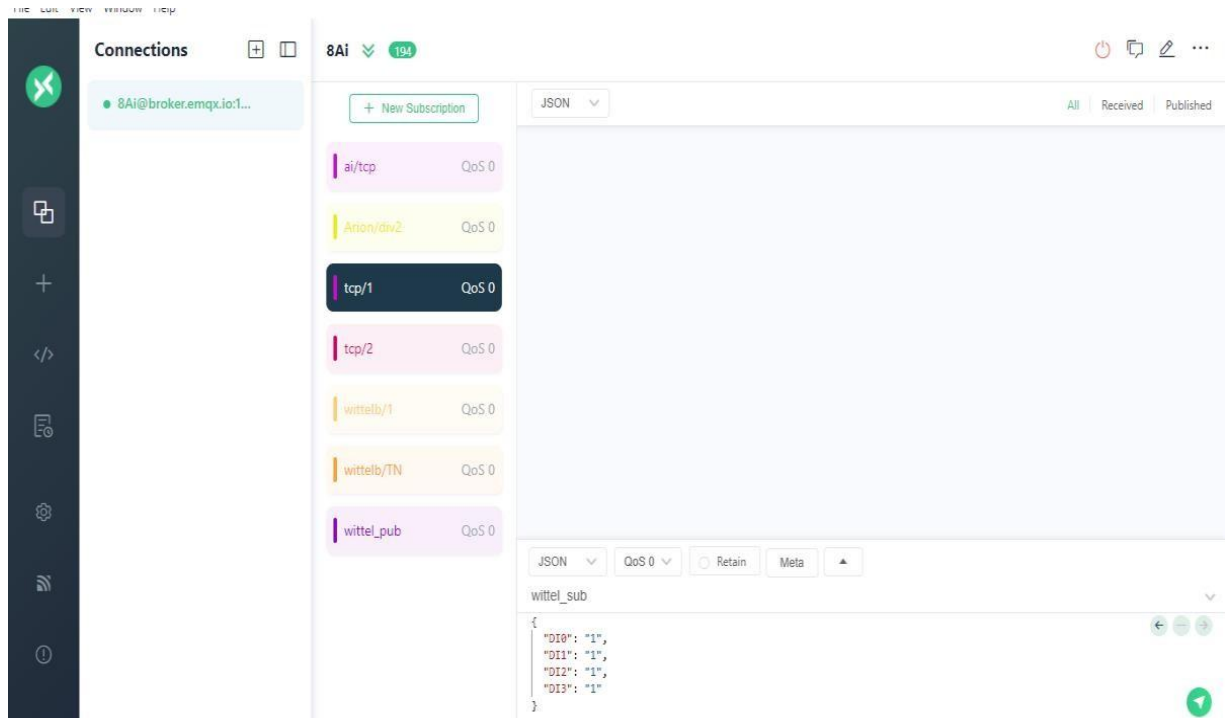
When the device sends analog input data, it is formatted as a JSON payload as shown below:

```
{  
  "AI0": "1023",  
  "AI1": "980",  
  "AI2": "450",  
  "AI3": "0"  
}
```

- Each key (AI0, AI1, etc.) represents an **analog input channel**.
- The corresponding value represents the **analog reading** (e.g., **0–1023** for a 10-bit ADC or **0–4095** for a 12-bit ADC).

Note:

You can visualize these analog values on your MQTT dashboard, broker console, or cloud platform by subscribing to the same topic where the device publishes data.

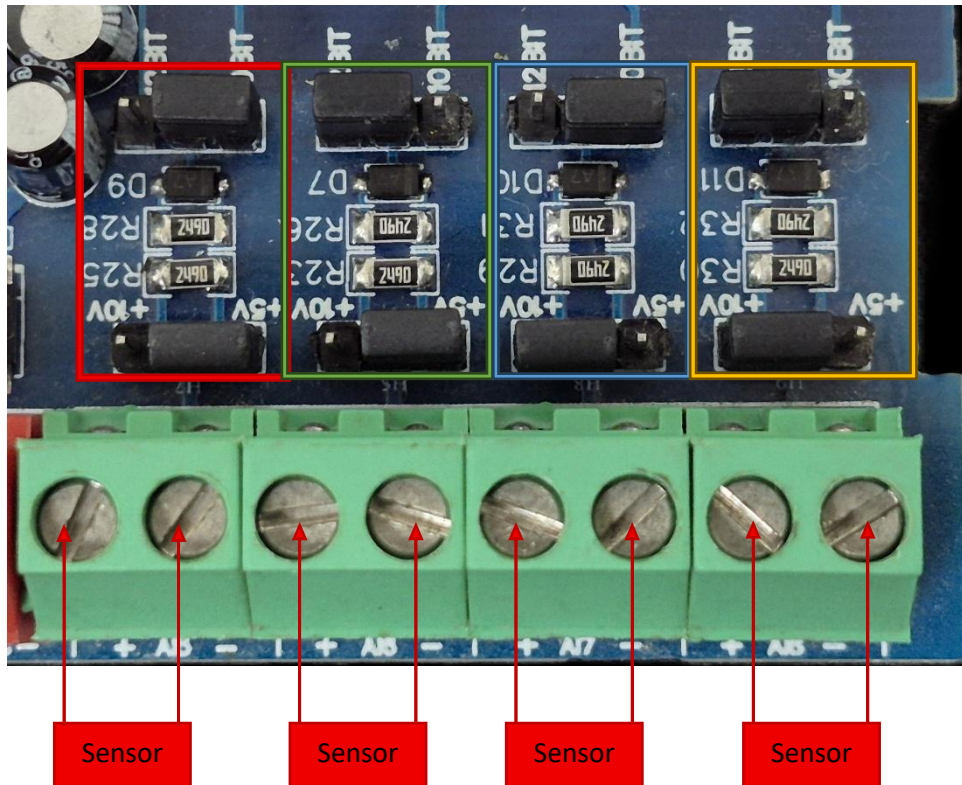


The screenshot displays the wittelB MQTT client interface. On the left, a dark sidebar contains navigation icons. The main window is divided into three panels:

- Connections:** Shows a single connection to `8Ai@broker.emqx.io:1883` with a status of "Online".
- Subscriptions:** A list of subscriptions with their respective QoS values:
 - `ai/tcp` (QoS 0)
 - `Antony/dw2` (QoS 0)
 - `tcp/1` (QoS 0)
 - `tcp/2` (QoS 0)
 - `wittelb/1` (QoS 0)
 - `wittelb/TN` (QoS 0)
 - `wittel_pub` (QoS 0)
- Message View:** Displays a message received on the `wittel_sub` topic. The message is a JSON object:

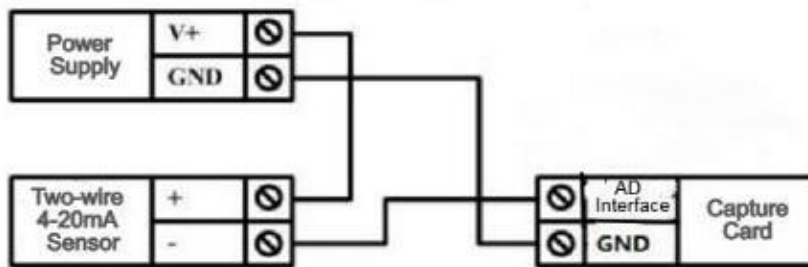
```
{  "DI0": "1",  "DI1": "1",  "DI2": "1",  "DI3": "1"}
```


AI

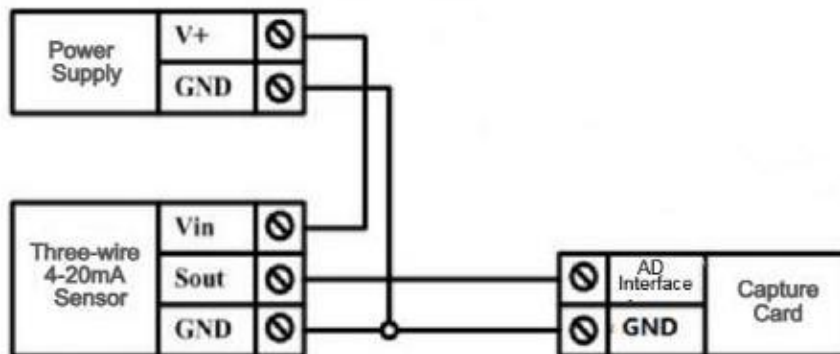


- **Analog Input (AI) Operation:** Each channel provides two terminals: + (Positive) and - (Negative/Common). These are used to read external analog sensor signals.
- **Channel Configuration (Visual Guide):** Each of the four AI channels features two dedicated jumpers to independently set the voltage range and bit resolution. Refer to the color-coded boxes in the board diagram to configure your desired scenario:
 - **Red Box (5V, 10-bit):** Set the voltage jumper to **+5V** and the resolution jumper to **10 BIT**.
 - **Green Box (5V, 12-bit):** Set the voltage jumper to **+5V** and the resolution jumper to **12 BIT**.
 - **Blue Box (10V, 10-bit):** Set the voltage jumper to **+10V** and the resolution jumper to **10 BIT**.
 - **Yellow Box (10V, 12-bit):** Set the voltage jumper to **+10V** and the resolution jumper to **12 BIT**.
- **Sensor Wiring Connections:**
 - The analog sensor's positive signal wire should be connected to the **+** terminal of the desired channel.
 - The sensor's common or ground wire should be connected to the corresponding **-** terminal.
- **Crucial Setup Recommendation:** Always ensure that the voltage jumper setting strictly matches the maximum output range of your connected sensor (either 5V or 10V) to guarantee accurate analog-to-digital conversion and prevent data clipping.

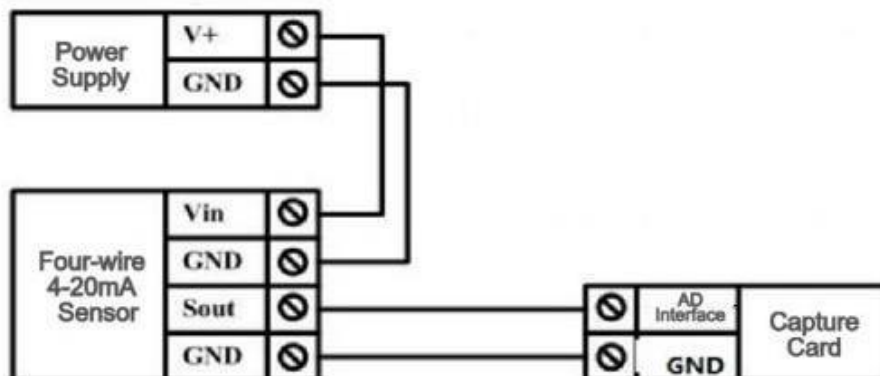
Two-wire Sensor Wiring Diagram

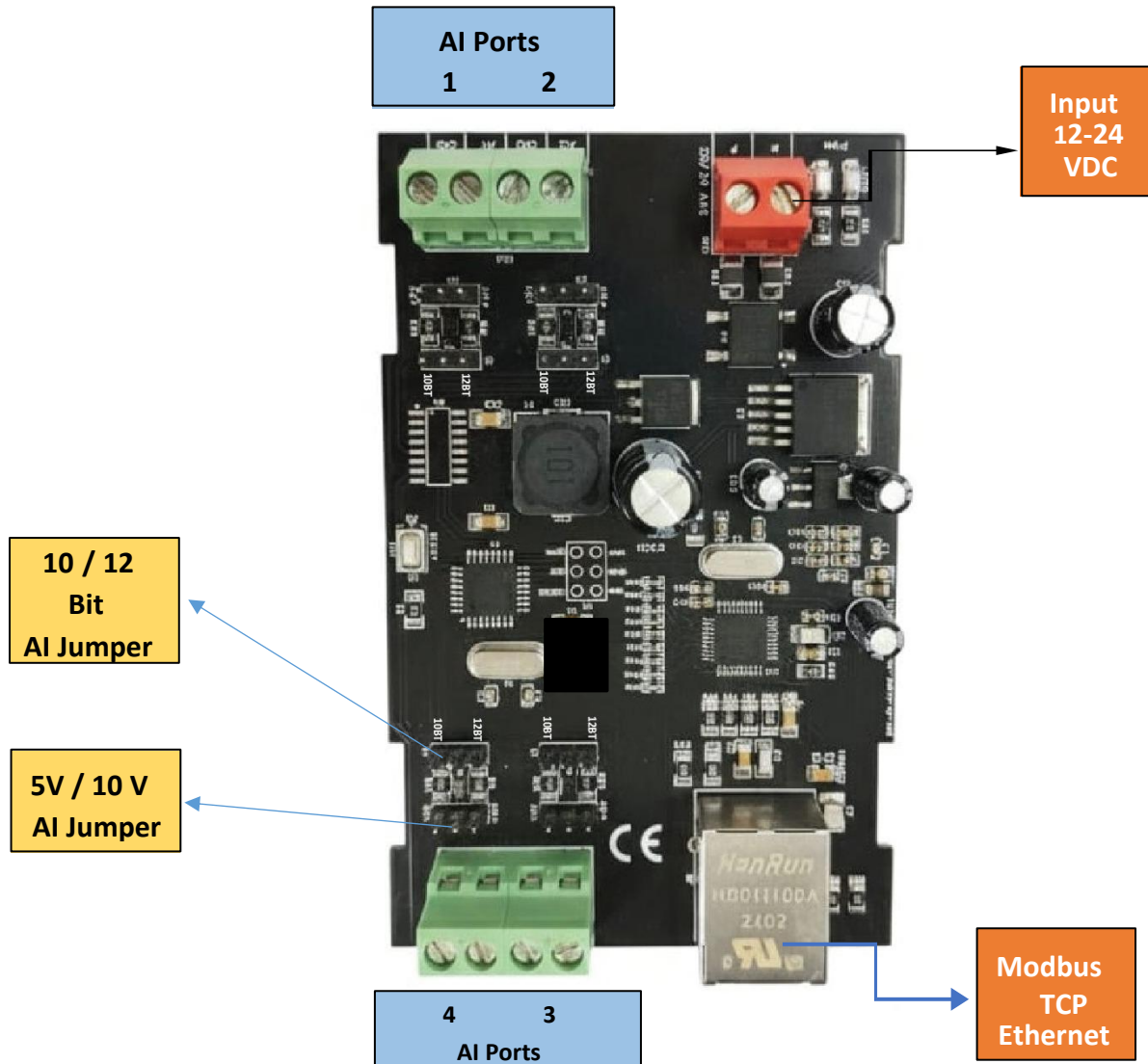


Three-wire Sensor Wiring Diagram



Four-wire 4-20mA Sensor





Contact us: -

Augmatic Technologies Pvt. Ltd.,
Plot no 6, Shah Industrial Estate II,
Kotambi,
Vadodara – 391510.

Email – Sales@wittelb.com

Support: support@wittelb.raiseaticket.com / +91 94038 92805